STAT2402: Week 1A Computer Laboratory

Getting Started:
Log in at one of the PCs and start up the software package RStudio. If you have problems either logging in or starting RStudio, ask the lab demonstrator for help.

When typing in R commands in the console window you can use the arrow keys to speed things up. The 'up' arrow gives you the previous command that you typed. The usual prompt sign for R is `>`. If you get a `+` prompt sign instead, it means that R is awaiting the completion of the previous command that you typed in. This can happen because you have forgotten to close parentheses, for instance. Just type in the remainder of the command and press enter.

You can ask the lab demonstrator for help at any point when you have a problem.

This laboratory session covers the following topics:

1. Probability.
2. Random variables.

Exercise 1: Binomial Problems

1. On Pingelap Island, 10% of the population is colour blind. A researcher selects 50 people at random from the island. Let the random variable X denote the number of people, out of the 50, who are colour blind.

   (a) State the distribution of the random variable X.
   Solution:
   $X \sim \text{Bin}(50, 0.10)$.

   (b) Determine the probability that of these 50 people, exactly 7 are colour blind.
   Solution: $\text{P}(X = 7) = 0.1076$ (4 d.p., from R).

   (c) Determine also the following:
   i. $\text{P}(X = 4)$; Solution: 0.1809 (4 d.p., from R)
   ii. $\text{P}(X \leq 6)$. Solution:0.7702 (4 d.p., from R)

   (d) What are the mean and variance of the random variable X?
   Solution: Since X is a Binomial random variable,

$$\begin{aligned} \mathbb{E}[X] &= np = 50 \times 0.1 = 5 \\ \text{Var}(X) &= np(1-p) = 50 \times 0.1 \times 0.9 = 4.5. \end{aligned}$$

2. Let X be the number of heads from 10 tosses of a fair coin. Evaluate the following probabilities:

   (a) $\text{P}(X = 5)$; Solution: From Tables, $\text{P}(X = 5) = 0.2461$ (4 d.p.). Alternatively,

$$\text{P}(X = 5) = \binom{10}{5} 0.5^5 (1 - 0.5)^{10-5} = 0.2461 \text{ (4 d.p.)}.$$

   (b) $\text{P}(X > 7)$; Solution: $\text{P}(X > 7) = 1 - \text{P}(X \leq 7) = 1 - 0.9453 = 0.0547$ (4 d.p., Tables).

   (c) $\text{P}(3 \leq X \leq 8)$. Solution: $\text{P}(3 \leq X \leq 8) = \text{P}(X \leq 8) - \text{P}(X < 3) = \text{P}(X \leq 8) - \text{P}(X \leq 2) = 0.9893 - 0.0547 = 0.9346$ (4 d.p., from Tables).

3. A botanist researching flower bulbs knows that 90% of its bulbs will flower. They are sold in packets of 12 randomly selected bulbs with a guarantee that the packet will be replaced if 100% flowering is not achieved.

(a) What is the probability that it will be necessary to replace a given packet under this guarantee? Interpret this probability.

(b) What would be the probability of replacing a packet if the guarantee covered only at least 10 out of 12 bulbs flowering? Comment on your findings in parts (a) and (b).

Solution: Let X denote the number of bulbs flowering out of 12. Then $X \sim \text{Bin}(12, 0.9)$.

(a) $P(X \leq 11) = 1 - \binom{12}{12}(0.9)^{12}(0.1)^0 = 0.7175$ (by calculation or R).
In the long run, 71.75% of the packets are replaced (surely the business will not survive!).

(b) $P(X \leq 9) = 1 - \binom{12}{12}(0.9)^{12}(0.1)^0 - \binom{12}{11}(0.9)^{11}(0.1)^1 - \binom{12}{10}(0.9)^{10}(0.1)^2 = 0.1109$.
In the long run, 11.09% of the packets are replaced (still very high!).

Exercise 2: Poisson Problems

1. While John is in his office, he receives 4 phone calls per hour on average. Assume that the number of calls within any interval of time is Poisson distributed.

   (a) What is the probability that the phone rings at least 4 times between 10am and 11am? Solution: Let X be the number of calls received during one hour. Then X is a Poisson random variable with parameter $4 \times 1 = 4$, and the required probability is

   $$\begin{aligned} \text{P}(X \geq 4) &= 1 - \text{P}(X \leq 3) \\ &= 1 - 0.4335 \qquad \text{(from Tables)} \\ &= 0.5665. \end{aligned}$$

   (b) If John takes a 30 minute lunch break, what is the probability that the phone does not ring during that time? Solution: 30 minutes = 0.5 hours; the number of calls received in any 30 minute interval is a Poisson random variable with parameter $4 \times 0.5 = 2$. Letting $X \sim \text{Poisson}(2)$, the required probability is $\text{P}(X = 0)$:

   $$\text{P}(X = 0) = e^{-2} \cdot \frac{2^0}{0!} = e^{-2} = 0.1353 \text{ (4 d.p.).}$$

   (c) What is the expected number of times that the phone will ring during John's lunch break? What is the variance? Solution: $\mathbb{E}[X] = \text{var}(X) = 2$.

   (d) If John arrives at work at 9 am and leaves at 5 pm, what is the expected number of times that the phone will ring during the day? What is the variance? Solution: Since this is an 8 hour interval, the mean and variance are $\mathbb{E}[X] = \text{var}(X) = 4 \times 8 = 32$.

2. Hummingbirds arrive at a flower at a rate λ per hour.

   (a) How many visits are expected in x hours of observation?

   (b) What is the variance of the number of visits in x hours?

   (c) If significantly more variance is observed than expected, what might this tell you about hummingbird visits?

   Solution: Let X = number of visits in x hours. Then $X \sim Poisson(\lambda x)$

   (a) $\mathbb{E}(X) = \lambda x$

   (b) $\text{Var}(X) = \lambda x$

   (c) The mean and variance for a Poisson distribution are the same. If the variance is much larger than the mean then the distribution is unlikely to be Poisson.

3. Let $Y \sim \text{Poi}(6)$. Without using `R`, find:

   (a) $\text{P}(Y \geq 3)$; Solution: $\text{P}(Y \geq 3) = 1 - \text{P}(Y \leq 2) = 1 - 0.0620 = 0.9380$ (4 d.p., Tables).

   (b) $\text{P}(Y \leq 15)$; Solution: $\text{P}(Y \leq 15) = 0.9995$ (4 d.p., Tables).

   (c) $\text{P}(3 \leq Y \leq 15)$. Solution:

$$\begin{aligned} \text{P}(3 \leq Y \leq 15) &= \text{P}(\{Y \geq 3\} \cap \{Y \leq 15\}) \\ &= \text{P}(\{Y \geq 3\}) + \text{P}(\{Y \leq 15\}) - \text{P}(\{Y \geq 3\} \cup \{Y \leq 15\}) \\ &= 0.9380 + 0.9995 - 1 = 0.9375. \end{aligned}$$

4. Bacteria are spread across a plate at an average density of 1000 per square cm.

   (a) What is the chance of seeing no bacteria in the viewing field of a microscope if this viewing field is 4×10^{-4} square cm?

   (b) What is therefore the probability of seeing at least one bacterium cell?

   Solution: Let X = number of bacteria cells in a 4×10^{-4} square cm area. Then the mean number of bacteria in this area are $4 \times 10^{-4} \times 1000 = 4 \times 10^{-1} = 0.4$, and $X \sim \text{Pois}(0.4)$.

   (a) $P(X = 0) = \frac{e^{-0.4}}{0.4^0 \times 0!} = 0.6703$.

   (b) $P(X \geq 1) = 1 - P(X = 0) = 1 - 0.6703 = 0.3297$ (From `R`)

5. A firm collects large quantities of data. Occasionally, typing errors cause data to be incorrectly entered. The number of typing errors per 20 pages of data is a Poisson random variable with mean 3.

   (a) What is the probability of there being 10 or more typing errors in 40 pages of data? Solution: Let X denote the number of typing errors in 40 pages of data. Then since $3 \times \frac{40}{20} = 6$, $X \sim \text{Poisson}(6)$. The required probability is

$$\text{P}(X \geq 10) = 1 - \text{P}(X \leq 9) = 1 - 0.9161 = 0.0839 \text{ (4 d.p.; Tables).}$$

   (b) What is the probability of there being between 5 and 9 (inclusive) typing errors in 40 pages of data? Solution: The required probability is

$$\begin{aligned} \text{P}(5 \leq X \leq 9) &= \text{P}(X \leq 9) - \text{P}(X \leq 4) \\ &= 0.9161 - 0.2851 = 0.6310 \text{ (4 d.p.; Tables).} \end{aligned}$$

   (c) What is the probability of there being less than 5 typing errors in 20 pages of data? Solution: Let Y be the number of typing errors in 20 pages of data; it is known that $Y \sim \text{Poisson}(3)$. The required probability is

$$\text{P}(Y < 5) = \text{P}(Y \leq 4) = 0.8153 \text{ (4 d.p.; Tables).}$$

   (d) What is the mean number of typing errors in 40 pages of data? What is the variance? Solution: The required mean and variance are $\mathbb{E}[X] = \text{var}(X) = 6$.

Reminder: Logging Off:
When you have finished, close down RStudio. Remember to log off from your computer before leaving.
